Abhishek Anand

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Current Position

Lamont-Doherty Earth Observatory, Columbia University New York City, NY
Postdoctoral Research Scientist, Advisor: Prof. Daniel Westervelt *November 2024 – Present*

Leveraging satellite remote sensing air pollution datasets, measurements from surface monitors, and ERA5 weather parameters to build ML-based algorithms for deriving high spatiotemporal resolution PM concentrations and health exposures in sub-Saharan Africa.

Education

Carnegie Mellon University Pittsburgh, PA
Doctor of Philosophy in Mechanical Engineering, Advisor: Prof. Albert Presto *May 2024*

Hong Kong University of Science and Technology Hong Kong
Master of Philosophy, Environmental Science, Policy and Management *August 2020*

Hong Kong University of Science and Technology Hong Kong
Master of Science, Environmental Engineering and Management *May 2017*

Indian Institute of Technology Delhi New Delhi, India
Bachelor of Technology in Civil Engineering *May 2015*

Past Research Experience

Carnegie Mellon University Pittsburgh, PA
Postdoctoral Research Associate, Advisor: Prof. Albert Presto *May 2024 – August 2024*

- Analyzed large datasets from instruments at the Pittsburgh site of Atmospheric Science and Chemistry mEasurement NeTwork (ASCENT), a network for high-time-resolution and long-term measurement in the U.S. for characterization of aerosol chemical composition.

Hong Kong University of Science and Technology Hong Kong
Research Assistantship, Advisor: Prof. Zhi Ning *June 2018 – August 2018*

- Impact of cross-sensitivity and environmental factors on performance of low-cost gaseous pollutant sensors. The sensors included Alphasense electrochemical gas sensors (CO, NO, NO_2, O_3 and SO_2) and NDIR (Non-Dispersive Infrared) CO_2 sensors.

Hong Kong University of Science and Technology Hong Kong
M.Sc. Student, Advisor: Prof. Irene Man Chi Lo *August 2016 – May 2017*

- Synthesis of visible-light-driven magnetic titanium oxide (TiO_2) - based nanophotocatalysts for degradation of persistent organic pollutants in wastewater.

Indian Institute of Technology Delhi New Delhi, India
Research Assistantship, Advisor: Prof. Saroj Kanta Mishra *June 2015 – July 2016*

- Analytical study of effects of geographical locations and sizes of mountains on the Indian Monsoon by simulating an Aqua planet on Community Atmosphere Model (v3.0).

Publications

Published

Anand, A., ..., Westervelt, D. M. Two decades of kilometer-scale daily $PM_{2.5}$ from satellite observations and machine learning reveal geographically diverging exposure in Ghana. *Atmospheric Measurement Techniques*. 2026.

Goldberg, P., **Anand, A.**, Goldberg, D., Westervelt, D. M. Attributing $PM_{2.5}$ and NO_2 Changes One Year After New York City's Congestion Pricing Policy. *Environmental Science & Technology Letters*. 2026

Westervelt, D. M., **Anand, A.**, et al. Nephelometer-based air sensor networks systematically underestimate $PM_{2.5}$ in dust-influenced West Africa. *npj Clean Air*. 2026.

Westervelt, D. M., Amooli, J. A., **Anand, A.** Twenty Years of High Spatiotemporal Resolution Estimates of Daily $PM_{2.5}$ in West Africa Using Satellite Data, Surface Monitors, and Machine Learning. *Environmental Science & Technology Air*. 2025.

Anand, A., Touré, N. D. E., Bahino, J., Gnamien, S., Hughes, A. F., Arku, R. E., ... & Presto, A. A. Low-cost hourly ambient black carbon measurements at multiple cities in Africa. *Environmental Science & Technology*. 2024.

Wei, P., Hao, S., Shi, Y., **Anand, A.**, et al. Combining Google traffic map with deep learning model to predict street-level traffic-related air pollutants in a complex urban environment. *Environment International*. 2024.

Anand, A., Kompalli, S., Ajiboye, E., & Presto, A. A. Estimation of hourly black carbon aerosol concentrations from glass fiber filter tapes using image reflectance-based method. *Environmental Science: Atmospheres*. 2023.

Wei, P., Brimblecombe, P., Yang, F., **Anand, A.**, et al. Determination of local traffic emission and non-local background source contribution to on-road air pollution using fixed-route mobile air sensor network. *Environmental Pollution*. 2021.

Anand, A., Wei, P., et al. Protocol development for real-time ship fuel sulfur content determination using drone-based plume sniffing microsensor system. *Science of the Total Environment*. 2020.

Wei, P., Sun, L., **Anand, A.**, Zhang, Q., et al. Development and evaluation of a robust temperature sensitive algorithm for long term NO_2 gas sensor network data correction. *Atmospheric Environment*. 2020.

Under review

Rogers, M. J., ..., **Anand, A.**, ..., Gentner, D. R. Observational constraints uncover the extensive contributions of biomass burning to organic particulate matter. (*Under review in Science*)

Amooli, J. A., **Anand, A.**, ..., Westervelt, D. M. Machine Learning-Based Prediction of Monthly Surface Ozone Over Sub-Saharan Africa Using Satellite-Derived Precursors, Meteorology, and Surface Measurements. (*Under review in ES&T Air*)

Pinakana, S., ..., **Anand, A.**, Westervelt, D. M. Size-Resolved Particle Mass and Number Performance Evaluation of Five Consumer Grade Air Sensors in New York. (*Under review in ES&T Air*)

Cheng, K., ..., **Anand, A.**, ..., Westervelt, D. M., Asante, Kwaku Poku. Chemical Composition and Source Apportionment of $PM_{2.5}$ Personal Exposure to Women and Children in Rural Ghana. (*Under review in ES&T Air*)

Research Data Products

Anand, A., et al. (2026). *Gridded Disease-Specific $PM_{2.5}$-Attributable Mortality in Ghana by Age and Sex, 2005–2025* [Dataset].

Anand, A., Amooli, J. A., & Westervelt, D. (2026). *Gridded Daily $PM_{2.5}$ at 1 km × 1 km Resolution from 2005–2025: Ghana* (Version v2.0) [Dataset]. Zenodo. https://doi.org/10.5281/zenodo.19636051

Accepted Proposals

P20 CHART Center Pilot Projects Program (P3), Columbia University.

$40,000 **(Lead–PI)**. 1/1/2026–12/31/2026. Mapping the air pollution and heat stress data gap in Africa: low-cost source apportionment and high spatiotemporal resolution datasets.

Climate School Summer Internship Funding Program, Columbia University.

$6,000 **(Lead–PI)**. 6/1/2025–8/31/2025. Funding for hiring interns in Summer 2025 to work on projects related to climate, sustainable development and the environment.

Dowd Fellowship, Carnegie Mellon University.

$100,000 **(Lead–PI)**. 9/1/2022–8/31/2023. Covered my tuition and monthly stipend for one academic year during Ph.D.

Fellowships and Awards

Jane Warren Award, Health Effects Institute 2026

1st Place, Hackathon on Applying Machine Learning for Subseasonal-to-Seasonal Climate Predictions, LEAP, Columbia University. 2025

Travel Grant recipient for American Association for Aerosol Research 2023 conference 2023

Philip and Marsha Dowd Fellowship, CMU (around $100,000 in tuition and stipend) 2022 – 2023

Milton Shaw Ph.D. Research Award, Department of Mechanical Engineering, CMU 2022

Postgraduate Studentship for the M.Phil. study at HKUST 2018 – 2020

HKUST awardee for the *8th Global Young Scientists Summit*, National Research Foundation, Prime Minister's Office, Singapore 2020

University Grants Committee Research Travel Grant, HKUST 2019

Hong Kong Government Innovation and Technology Fund Internship Award 2018

M.Sc. Excellent Student Scholarship, School of Engineering, HKUST 2017

Champion Award, BESTo Camp, HKUST Entrepreneurship Center 2017

Entrance Scholarship, School of Engineering, HKUST 2016

Ministry of Human Resources Development Scholarship at IIT Delhi, covering tuition fees for 4 years of undergraduate studies. 2011 – 2015

Invited Talks

UIUC-KYUSHU-NTU Trilateral Workshop, University of Illinois Urbana-Champaign *July 29, 2026*
Long-Term Spatiotemporal Trends in $PM_{2.5}$-Attributable Mortality and Socioeconomic Disparities in Ghana Using Satellite Measurements and Machine Learning.

NASA Goddard Space Flight Center (Aerocentre-CPC Seminar), Greenbelt, USA *May 2026*
Mapping Two Decades of High-Resolution $PM_{2.5}$ Exposure and Associated Health Impacts in Ghana Using Satellite Measurements and Machine Learning.

Health Effects Institute (HEI) Annual Conference, Chicago, USA *April 2026*
Mapping Two Decades of High-Resolution $PM_{2.5}$ Exposure and Health Impacts in Ghana Using Satellite Measurements and Machine Learning.

Climate and Health Evaluation for Adaptive Resilience (CLEAR), New York, USA *April 2026*
Long-Term Health Effects of $PM_{2.5}$ Exposure and Its Associated Socioeconomic Disparities in Ghana: Evidence from 21 Years of Spatiotemporal Data.

Kintampo Health Research Center (KHRC), Ghana Health Service, Kintampo, Ghana *January 2026*
Gridded Africa Surface Pollution Dataset (GRASP): Two Decades of Satellite-Derived Daily High-Resolution $PM_{2.5}$ Measurements in Ghana.

Environmental Protection Authority (EPA), Accra, Ghana *January 2026*
Gridded Africa Surface Pollution Dataset (GRASP): Two Decades of Satellite-Derived Daily High-Resolution $PM_{2.5}$ Measurements in Ghana.

GRAPHS Manuscript Series, Columbia University, NY *December 2025*
Mapping Two Decades of Daily High-Resolution $PM_{2.5}$ Data in Ghana Using Machine Learning.

Geochemistry Division, Columbia University, NY *October 2025*
Atmospheric Black Carbon Measurements by Applying Image Processing Method on Filter Tapes.

Department of Civil Engineering, University of Illinois Urbana-Champaign, IL *October 2025*
Leveraging Satellite Measurements, Surface Monitors, and Machine Learning for Generating 20 Years of High-Resolution Daily $PM_{2.5}$ in Ghana.

SPARTAN & CAMS-Net Joint Meeting, Washington University in St. Louis, MO *June 2025*
Two Decades of High-Resolution Daily $PM_{2.5}$ in Ghana: A Machine Learning Approach.

Lamont 75th Mini-Symposium: The Data Drive Discovery, Columbia University, NY *May 2025*
Leveraging Satellite Measurements to Build Machine Learning Models for Estimating 20 years of High-Resolution

Gridded Daily $PM_{2.5}$ for Ghana.

Air Sensors International Conference, Riverside, CA *October 2024*
Low-cost methods for measurement of $PM_{2.5}$ composition at African cities by exploiting existing Beta Attenuation Monitors.

Conference Presentations

Gentner, D. R., ..., **Anand, A.**, et al. Examining the effects of increased temperatures and wildfire smoke on urban air quality in New York City. *American Geophysical Union Fall Meeting*, San Francisco, CA. 2026.

Amooli, J. A., **Anand, A.**, et al. Estimating Monthly Surface Ozone Across Sub-Saharan Africa from Satellite Observations and Meteorology Using Machine Learning. *AGU Fall Meeting 2026*, San Francisco, CA, USA. 2026.

Anand, A., Adabouk Amooli, J., Westervelt, D. M. Mapping Two Decades of Daily High-Resolution $PM_{2.5}$ Exposure in Ghana Using Satellite Observations and Machine Learning. *Health Effects Institute (HEI) Annual Conference*, Chicago, IL. Jane Warren Award Presentation. 2026.

Gentner, D. R., Rogers, M. R., Joo, T., ..., **Anand, A.**, et al. Top-down constraints on the contributions of biomass burning to atmospheric particulate matter. *109th Canadian Chemistry Conference & 76th Canadian Chemical Engineering Conference*, Toronto, ON, Canada. 2026.

Gentner, D. R., Rogers, M., Joo, T., ..., **Anand, A.**, et al. Examining the impacts of climate change and biomass burning on air quality in the 21st century. *Goldschmidt Conference*, Montreal, QC, Canada. 2026.

Anand, A., Adabouk Amooli, J., Westervelt, D. M. Tracking Long-Term Air Pollution and Health Exposure in Ghana: Twenty Years of High-Resolution $PM_{2.5}$ Maps Using Satellite Data, Surface Observations, and Machine Learning. *American Geophysical Union Fall Meeting*, New Orleans, LA. 2025.

Westervelt, D. M., Goldberg, P., **Anand, A.** Spatial and Distributional Air Quality Impacts of New York City's Congestion Pricing Policy: Evidence from Ground and Satellite Observations. *American Geophysical Union Fall Meeting*, New Orleans, LA. 2025.

Anand, A., Amooli, J. A., Westervelt, D. M. Leveraging Satellite Measurements, Surface Monitors, and Machine Learning for Estimating 20 Years of High-Resolution Gridded $PM_{2.5}$ in Ghana. *American Association for Aerosol Research*, Buffalo, NY. 2025.

Anand, A., Opinde, G., Mwendwa, T. M., Habineza, T., Presto, A., DeCarlo, P. F., Nault, B. A., Westervelt, D. M. Aerosol Chemical Composition at an Urban Core Location in Nairobi, Kenya. *American Association for Aerosol Research*, Buffalo, NY. 2025.

Zeng, Z., **Anand, A.**, Habineza, T., Bahreini, R., Dillner, A. M., Russell, A. G., Ng, N. L., Jen, C., Presto, A. High-Resolution $PM_{2.5}$ Metals in Pittsburgh: Trends, Fireworks, and Wildfire Impacts from the ASCENT Network. *American Association for Aerosol Research*, Buffalo, NY. 2025.

Rogers, M., Joo, T., Zhang, L., Hass-Mitchell, T., Murphy, B., Zeng, Z., Habineza, T., **Anand, A.**, et al. Observational Constraints on the Total Contributions of Biomass Burning to Ambient Organic Aerosol. *American Association for Aerosol Research*, Buffalo, NY. 2025.

Anand, A., Presto, A., et al. Estimation of Total and Biomass-Based BC at African Cities by Applying Image-Reflectance Method on BAM Tapes. *American Association for Aerosol Research*, Albuquerque, NM. 2024.

Anand, A., Presto, A., Farimani, A. B. Development of an improved deep learning-based $PM_{2.5}$ model for predicting high pollution episodes in Pittsburgh by leveraging GEOS-CF atmospheric composition data. *American Geophysical Union*, San Francisco, CA. 2023.

Anand, A., Presto, A., Farimani, A. B. Developing a machine learning-based daily $PM_{2.5}$ forecast model with GEOS-CF and land use parameters. *American Association for Aerosol Research*, Portland, OR. 2023.

Anand, A., Kompalli, S. P., et al. Black carbon measurements in multiple cities of sub-Saharan Africa with low-cost image reflectance method. *American Association for Aerosol Research*, Portland, OR. 2023.

Anand, A., Presto, A., Kompalli, S. P., et al. Hourly black carbon measurements in Africa using cell phone camera images. *American Geophysical Union*, Chicago, IL. 2022.

Anand, A., Presto, A., Kompalli, S. P., et al. Low-Cost black carbon detection from Beta Attenuation Monitors using image reflectance-based method. *American Association for Aerosol Research*, Raleigh, NC. 2022.

Kim, S., **Anand, A.**, Rajan, P. E., Presto, A. Comparison of organic aerosol composition and source distributions across different urban environments. *American Association for Aerosol Research*, Raleigh, NC. 2022.

Anand, A., Presto, A., Kompalli, S. P., et al. Estimation of hourly BC from BAM tapes using image reflectance-based method. Air Sensors International Conference, Pasadena, CA. 2022.

Anand, A., Gali, N. K., Yang, F., et al. Laboratory calibration, validation and protocol development to use UAV borne sensor system for fuel sulfur content-based field screening of OGVs. *Global Young Scientists Summit*, Singapore. 2020.

Anand, A., Gali, N. K., Westerdahl, et al. Technology development and evaluation of an ultra-compact ship fuel Sulfur sniffing sensor system. *Freight and Environment: Ports of Entry*, Newark, NJ. 2019.

Ning, Z., **Anand, A.**, Gali, N. K., et al. Protocol Development of using Sniffing Method to Identify High Sulfur Fueled Ships. *Freight and Environment: Ports of Entry*, Newark, NJ. 2019.

Teaching Experience

Guest Lecture, Computing and Research Methods for Climate Data Science, Columbia University (*February 2026*)

Introduced major climate datasets, tools to work with them, data structures, and research applications.

Guest Lecture, Air Pollution & Measuring the Environment, Columbia University (*November 2025*)

Delivered lecture on remote sensing principles and practical applications of NASA and ESA satellite-derived air pollution datasets.

Future Faculty Career Program, Carnegie Mellon University 2020 – 2024

Designed to help early career researchers develop their teaching skills for a faculty career.

Teaching Assistant

Renewable Energy Engineering – 24792, CMU	Spring 2023
Fluid Mechanics – 24231, CMU	Spring 2022
GIS for Environmental Professionals – EVSM5240, HKUST	Fall 2019
Carbon Emission Trading – ENVR6090A, HKUST	Spring 2019

Peer Tutor for Undergraduate Students, CMU 2022 – 2023

Physics I for Science Students (33121), Physics II for Biological Sciences and Chemistry Students (33122), Physics I for Engineering Students (33141), Physics II for Engineering and Physics Students (33142), Calculus (21111-122), Differential Equations (21260)

Workshops and Training

Lead Instructor, Climate Change Workshop, PATS MECOR Africa Meeting *October 4 – 9, 2026*

Working with Spatial Environmental Exposure Data for Epidemiological Research: From Air Pollution to Heat Exposure, Brackenhurst, Kenya.

Training for epidemiological researchers on gridded environmental exposure datasets, Python/Google Colab, satellite and reanalysis data, $PM_{2.5}$ exposure mapping, and preparation of exposure datasets for epidemiological analysis.

Co-lead Instructor, Workshop on Satellite Data Applications and Innovations in Air Quality Research and Health Applications, Ghana

Kintampo Health Research Center (KHRC), Kintampo, Ghana *January 15–17, 2026*
Environmental Protection Authority (EPA), Accra, Ghana *January 12–14, 2026*

Training included Python for environmental data analysis; fundamentals of satellite remote sensing; accessing and downloading satellite-derived products from NASA and ESA, including TROPOMI, OMI, and MODIS; accessing and processing meteorological and spatial environmental datasets from ECMWF, including ERA5, using NetCDF, HDF4/5, xarray, pyhdf, and h5py; applications of these datasets to air-quality analysis and exposure assessment; and using AirQ+ and MR–BRT methods to estimate the health burden associated with $PM_{2.5}$ exposure.

Instructor, Barnard College, Columbia University, New York, USA *September 2025*

Workshop on satellite data analysis using MODIS Aerosol Optical Depth (AOD), including methods for relating $PM_{2.5}$ to AOD and comparing air quality relationships across multiple U.S. cities.

Advising Experience

Graduate Research Mentor

Sai Pinakana - Ph.D. student, DEES, Columbia University — Fall 2025 – Present
Sizhou Su - Master's student, Columbia University — Summer 2025 – Present
Aziz Bhetasiwala - Master's student, Mechanical Engineering, CMU — Fall 2023 – Summer 2024
Ria Sharma - Master's student, Mechanical Engineering, CMU — Fall 2023

Undergraduate Research Mentor

Polina Goldberg - Undergraduate student, Data Science, Columbia University — Summer 2025 – 2026
Elsevar Zeynalov - Undergraduate student, Data Science, Columbia University — Summer 2025
Ria Sharma - Undergraduate student, Mechanical Engineering, CMU — Summer 2023
Jordan Petzold - Undergraduate student, Mechanical Engineering, CMU — Summer 2023
Jocelyn Kiefel - Undergraduate student, Mechanical Engineering, CMU — Summer 2023
Shaborn Leggette - Undergraduate student, Mechanical Engineering, CMU — Summer 2023
Max Labovitz - Undergraduate student, Mechanical Engineering, CMU — Summer 2022

Academic and Professional Service

Journal Peer Reviewer — 2023 – Present

Geoscientific Model Development, Atmospheric Chemistry & Physics, Environmental Science & Technology, Atmospheric Measurement Techniques, Environmental Science & Technology Air, Environment International, Environmental Monitoring and Assessment, Environmental Science and Pollution Research, and Scientific Reports.

Member, AGU Atmospheric Science Section Early Career Committee — January 2026 – Present

Core Representative, Postdoc/ARS Hardship Support Fund Committee, Columbia University — 2025 – Present

Panelist and Reviewer — September 2026

NASA Early Career Investigator Program in Earth Science (ECIP-ES), Earth Science Data Systems panel.

Co-convener, American Geophysical Union Fall Meeting 2026 — December 2026

Town Hall: *A Discussion with Program Managers for Early-Career Scientists.*

Co-convener, American Geophysical Union Fall Meeting 2026 — December 2026

A046: *Atmospheric Sciences Student, Postdoc, and Early Career Flash Talks.*

Co-convener, American Geophysical Union Fall Meeting 2026 — December 2026

AGU26 *Early Career Mini-Workshop.*

Coordinator, Ocean and Climate Physics Department Seminar, Columbia University — 2025 – 2026

Scientist-in-Residence, New York Academy of Sciences — 2025 – 2026

United Charter High School - Advanced Math and Science, Bronx, New York City.

Collaborated with a teacher to develop and deliver modules on air pollution sources and monitoring for high school students from underserved communities in the Bronx. Designed student research projects involving the development of an air purification system and a sensor module to measure particulate removal efficiency.

Session Chair, American Association for Aerosol Research Conference 2025 — October 2025

Advancing Aerosol Science Through Data Analysis session, Buffalo, NY, USA.

Panelist, SPARTAN & CAMS-Net Meeting — June 2025

Low-Cost Monitoring of Atmospheric Particulate Matter, Washington University in St. Louis, MO, USA.

President, AAAR Student Chapter, Carnegie Mellon University — 2023 – 2024

Coordinator, Center for Atmospheric Particle Studies Seminar, Carnegie Mellon University — 2022 – 2023

Core Committee Member, CAPS Laboratory, Carnegie Mellon University 2021 – 2022

Relevant Coursework

Probability, Machine Learning and Statistics, Carnegie Mellon University
Introduction to Deep Learning for Engineers, Intermediate Deep Learning, Machine Learning and Artificial Intelligence for Engineers, Probability and Estimation Methods for Engineering Systems, Statistical Learning and Modeling.

Air Quality and Atmospheric Sciences
Air Quality Engineering at CMU; Atmospheric Dynamics at HKUST, Numerical Simulation of Atmospheric Phenomena at IIT Delhi